Presentation Outline

Structured Rubric: 6 Core Academic Sections

I. Problem & Motivation

- The Reversal Curse in Transformer architectures.
- Why standard cross-encoders fail on inverse pairs.
- Pathological failure modes in directional NLI.

II. Corpus & Task Definition

- SemEval-2027 Task 2: DiCo-NLI & PhrasIS suite.
- 4 relational classes & deterministic reversal algebra.
- Official metrics: Weighted F_1, SoftCons, HardCons.

III. Methodology & Formulations

- Baseline Cross-Encoder vs. BSA Augmentation.
- Siamese Consistency Regularization Loss (CRL).
- Permutation involution operator $P \in \mathbb{R}^{4\times4}$.

IV. Symmetrized Decoding (SD) Proof

- Post-hoc equivariant projection in probability space.
- Mathematical proof guaranteeing SoftCons $\equiv$ 100%.
- Deterministic involutory tie-breaking rule.

V. Empirical Results & Analysis

- Benchmark comparison: 5 methods on Tracks 1 & 2.
- 4×4 Directional Transition Matrices & RCSI.
- Syntactic chunk breakdown (Nominal, Verbal, etc.).

VI. Conclusions & Takeaways

- Overcoming the Reversal Curse without retraining.
- Practical guidelines for DiCo-NLI submissions.
- Future directions in equivariant reasoning.